Morden MONOPOLY

Team 11

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https://github.com/Pollyyaooo/515_monopoly.git

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How might we reduce the administrative burden of tracking game state and enforcing rules in Monopoly, so that all players can focus on strategy and fun?

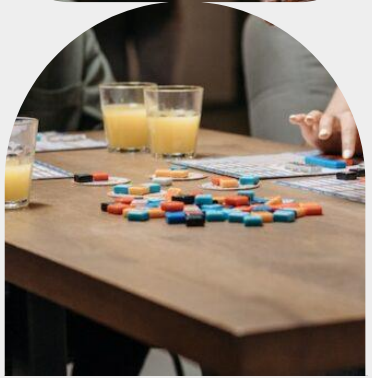
Why we want…

The problem

Monopoly requires players to manually track positions, property ownership, houses, rents, and card effects — creating cognitive overload, rule disputes, and slow gameplay.

Intended users

Anyone who plays Monopoly — from first-time players overwhelmed by rules, to experienced players who want faster, dispute-free games.



Solution

A smart Monopoly system that automatically tracks board state and enforces rules, so players never have to.

Board Intelligence

The physical board is enhanced with sensors to automatically detect and track:

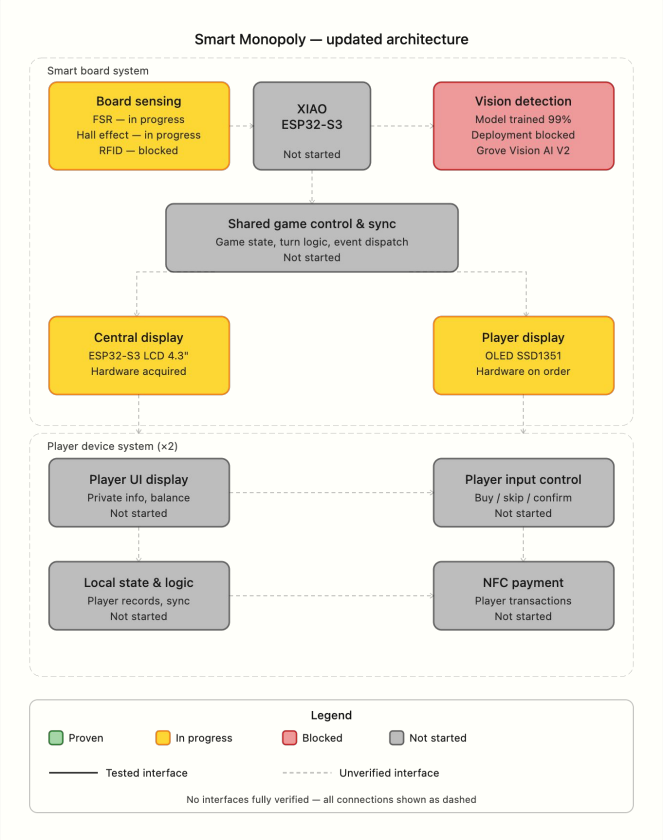
- Player token positions (who is where)
- Property ownership status
- Number and color of houses built on each property

Rules Intelligence

Game information is processed and displayed in real time across two interfaces:

- A Central Bank screen (public display) — shows card draws, rent calculations, property transactions…
- Individual player devices (private display) — shows personal balance, rent owed/collected, player-to-player transfers..

Architecture 2



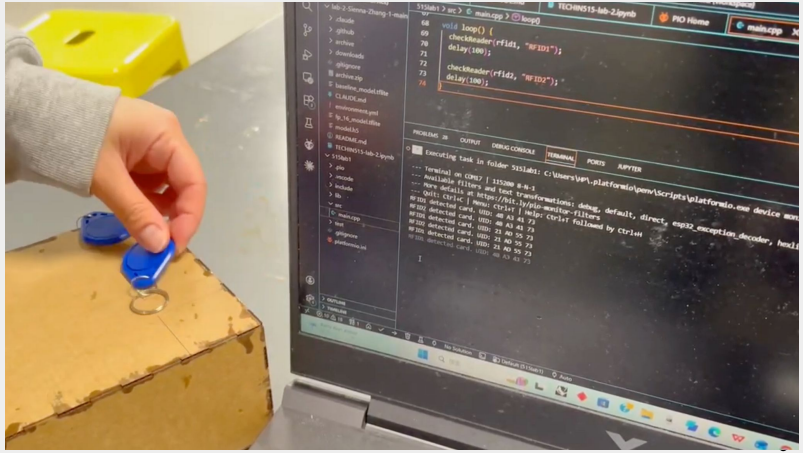
Validation 1 - RFID

We tested the RFID system with multiple tags and readers. A single RFID reader can successfully detect two tags simultaneously. However, when two RFID readers are placed too close to each other(<2cm), neither of them can detect any tags. This indicates signal interference between readers.

Prototype1_RFID_board_cover.MOV

Prototype1_RFID_1tag.MOV

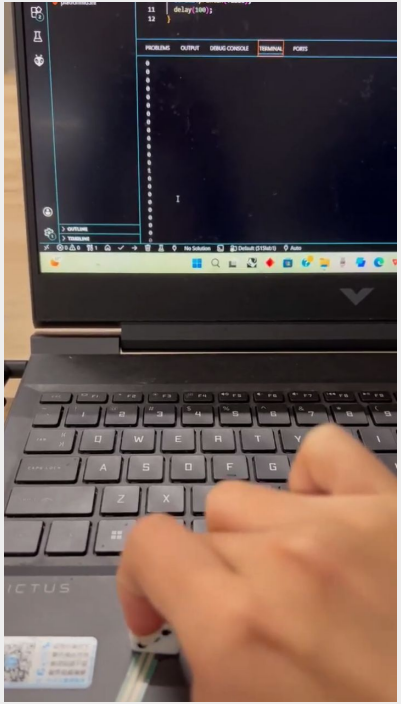
Prototype1_RFID_2tags.MOV



Validation 2 - FSR Sensor

We tested the force-sensitive resistor (FSR) for detecting object placement. We found that the contact area must be smaller than the circular sensing region of the FSR; otherwise, the sensor fails to detect pressure effectively. Additionally, very light objects are not reliably detected. Based on this, we are considering adding weights inside the game pieces to ensure sufficient pressure.

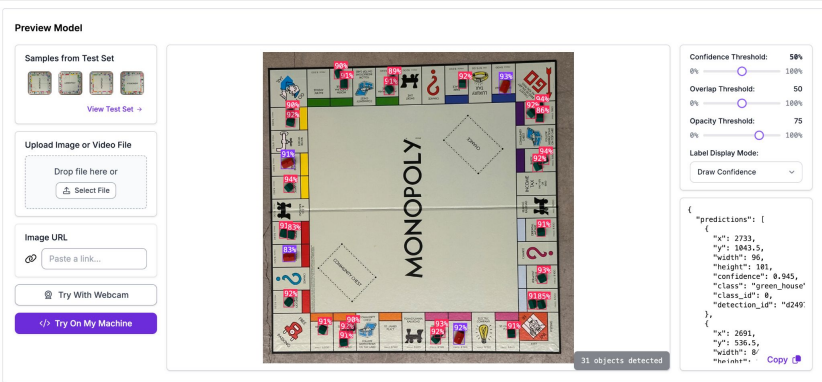
Prototype1_FSR.mp4



Validation 3 - Vision

We mounted the OV5647-62 camera on Grove Vision AI V2 and determined via live preview that a mounting height of approximately 65cm is required to fully capture the 50×50cm Monopoly board. We collected 67 annotated images of red and green houses, augmented to 161 images, and trained a YOLOv11 object detection model achieving mAP@50 = 99.0%, Precision = 99.9%, Recall = 98.7%. Qualitative webcam testing confirmed reliable color distinction between red and green houses across different board positions. However, deployment to Grove Vision AI V2 requires int8_vela.tflite format conversion, which has not yet been successfully completed.

https://app.roboflow.com/yitongs-workspace/monopoly-houses/models/monopoly-houses/1



Budget update

BUDGET SUMMARY									
Team Num	11	CFO Name:	Polly Yao	Total Budget ($):	$350.00	Total Spent ($):	$15.24	Remaining ($):	$334.76
#	Item Name	Quantity	Unit Price ($)	Total Price ($)	Supplier	Link	Notes	Instructor Approval	Actual ($)
1	A1324LUA-T	5	$1.30	$6.51	Digikey	https://www.digikey.com/en/products/detail/allegro-microsyste	Ordered 4.14.26. Arriving 5-8 business	approve	$15.24
2	Amphenol ICC (FCI) 609-2176-ND CONN	15	$0.74	$11.04	Digikey	https://www.digikey.com/en/products/detail/amphenol-icc-fci-/6		no approve	
3	waveshare 1.5inch RGB OLED Display Module	2	$21.93	$43.86	Amazon	https://www.amazon.com/dp/B07DBXMFSN?ref=fed_asin_title		no approve	
4	ALMOCN 3PCS PN532 NFC NXP RFID Module	1	$15.59	$15.59	Amazon	https://www.amazon.com/dp/B09VC5BQFC?ref=fed_asin_title		no approve	

Thank you!